

# Prototype of a Wireless Seismic Data Acquisition System

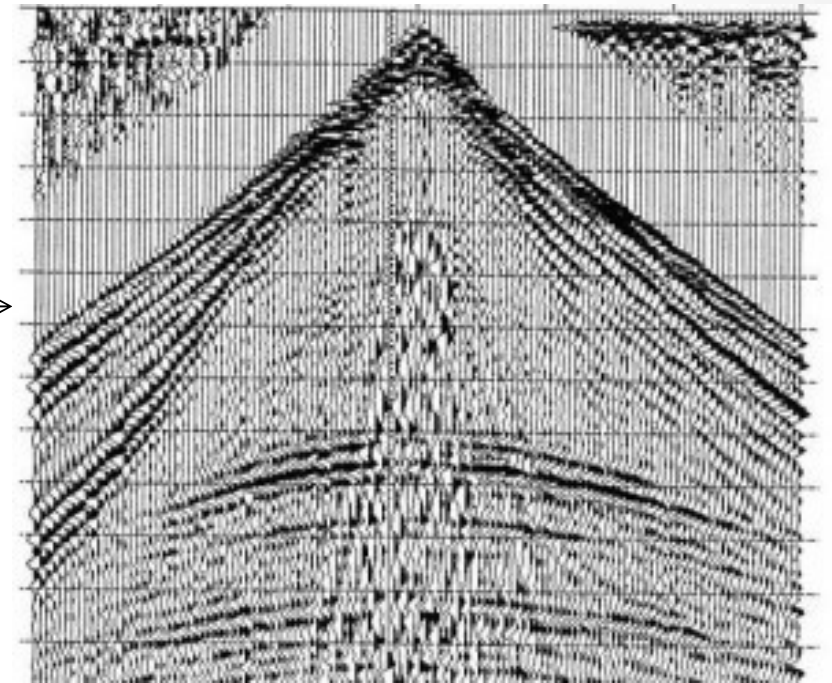
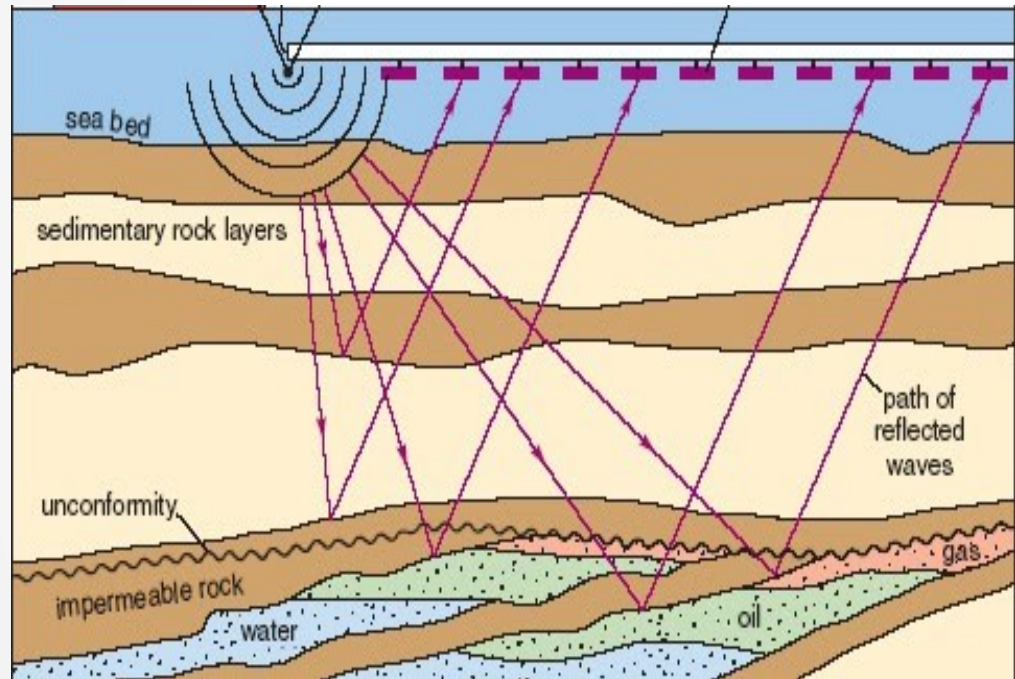
## Group Members

| Name                                      | ID        |
|-------------------------------------------|-----------|
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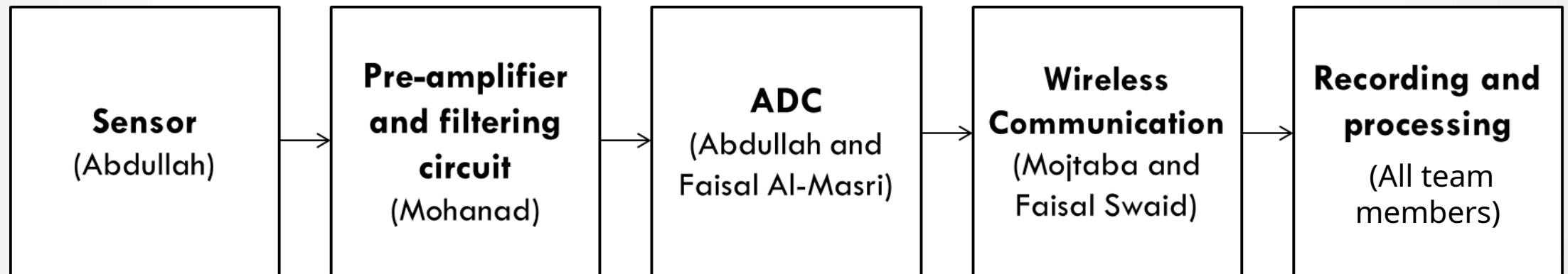
# OUTLINES

- Overview
  - Project Definition & Objectives
  - Project Management
  - Final Design
    - Sensors
    - Pre-ADC Conditioning Circuit
    - Wireless Communication
    - Signal Processing
- Design Verification
  - Conclusions
  - Recommendations
  - Demo

# Seismic Data ?



# Work Distribution

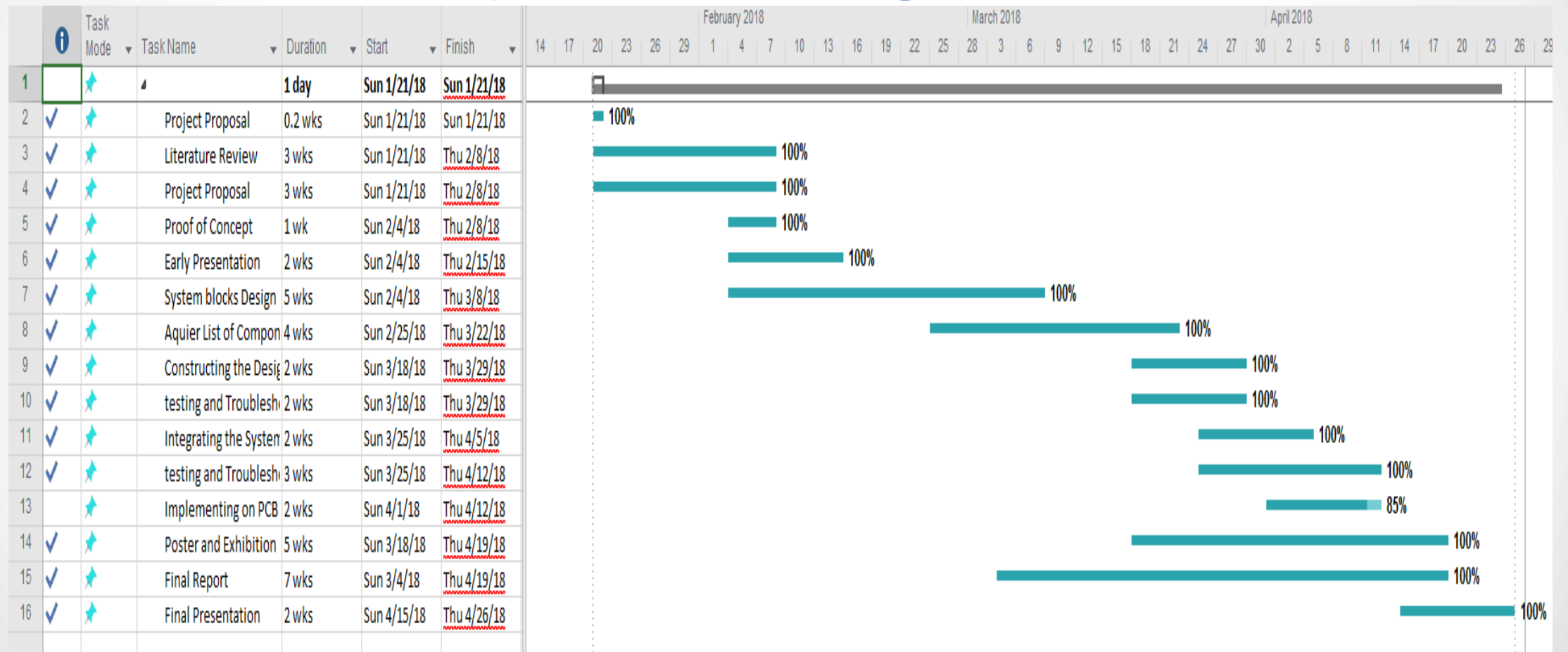


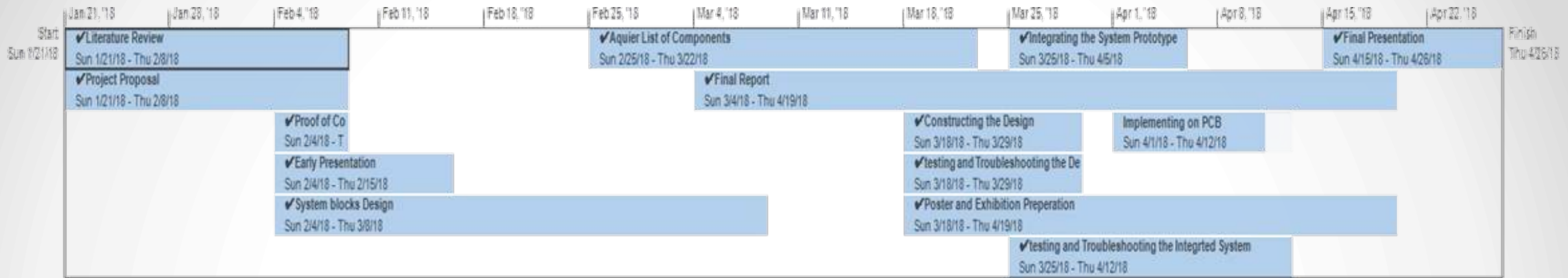
# Project Definition And Objectives

Implementing a complete wireless seismic data acquisition prototype and designing some parts of the system with emphasized low power consumption.

- Examine different possible seismic sensors.
- Implement the required conditioning circuits.
- Test different communication techniques (Serial & wireless).
- Build a wireless seismic data acquisition prototype with emphasizes on low power consumption.

# Project management





## PROJECT OVERVIEW

SUN 1/21/18 - THU 4/26/18

% COMPLETE

99%

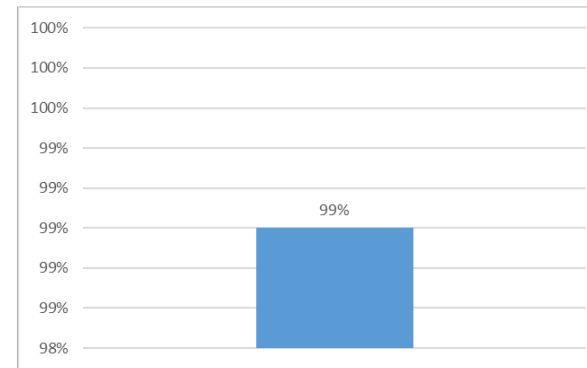
### MILESTONES DUE

Milestones that are coming soon.

| Name | Finish |
|------|--------|
|------|--------|

### % COMPLETE

Status for all top-level tasks. To see the status for subtasks, click on the chart and update the outline level in the Field List.



### LATE TASKS

Tasks that are past due.

| Name                | Start      | Finish      | Duration | % Complete | Resource Names |
|---------------------|------------|-------------|----------|------------|----------------|
| Implementing on PCB | Sun 4/1/18 | Thu 4/12/18 | 2 wks    | 85%        |                |

# Final Design

1. Sensors.
2. ADC.
3. Pre-ADC Conditioning.
4. Wireless Communication.
5. Signal Processing.

# Sensors

- Analog
  - Coil-Based Geophone.
  - Piezoelectric.
- Digital (MEMS)
  - Accelerometer.
  - Gyroscope.



Coil-Based  
Geophone

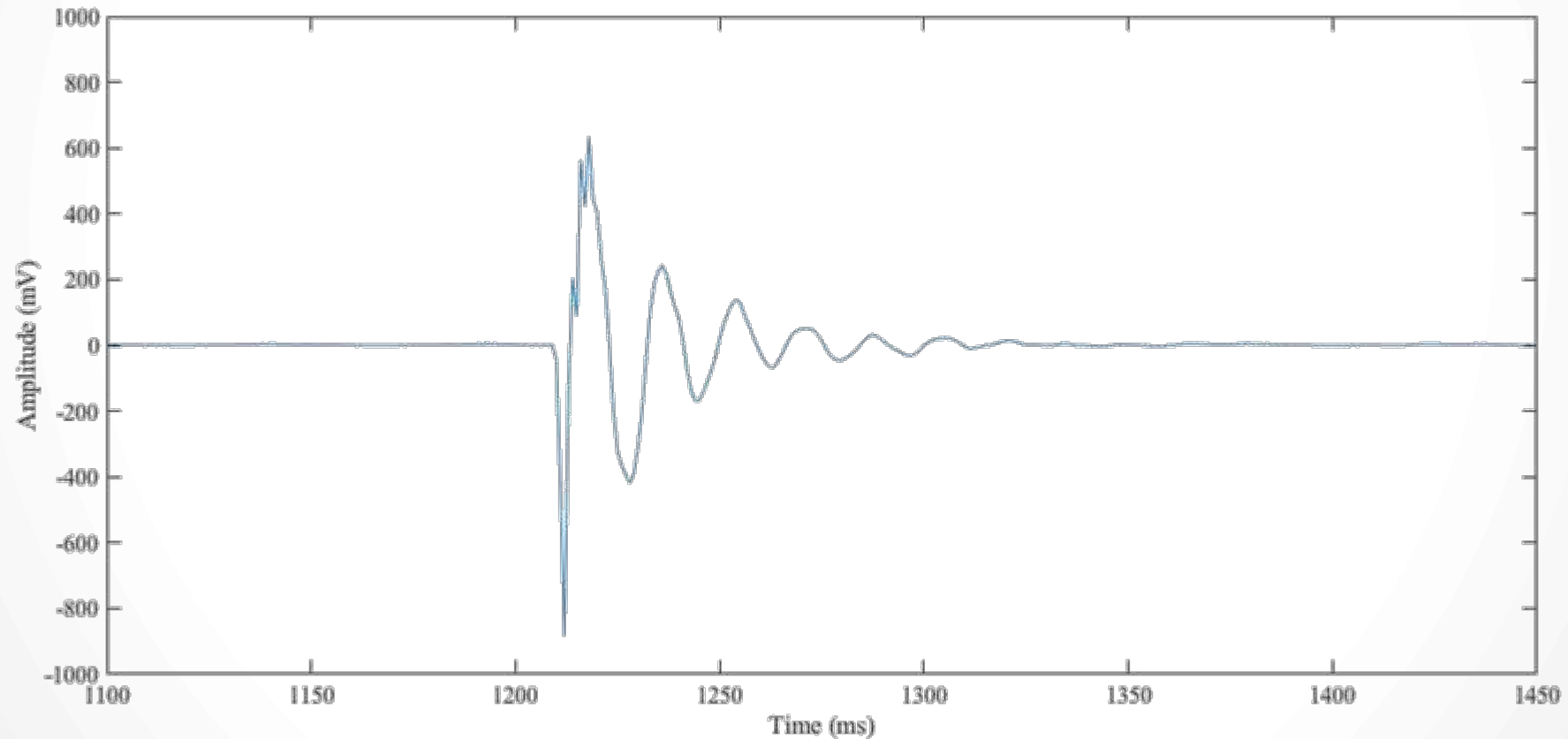


Piezoelectric

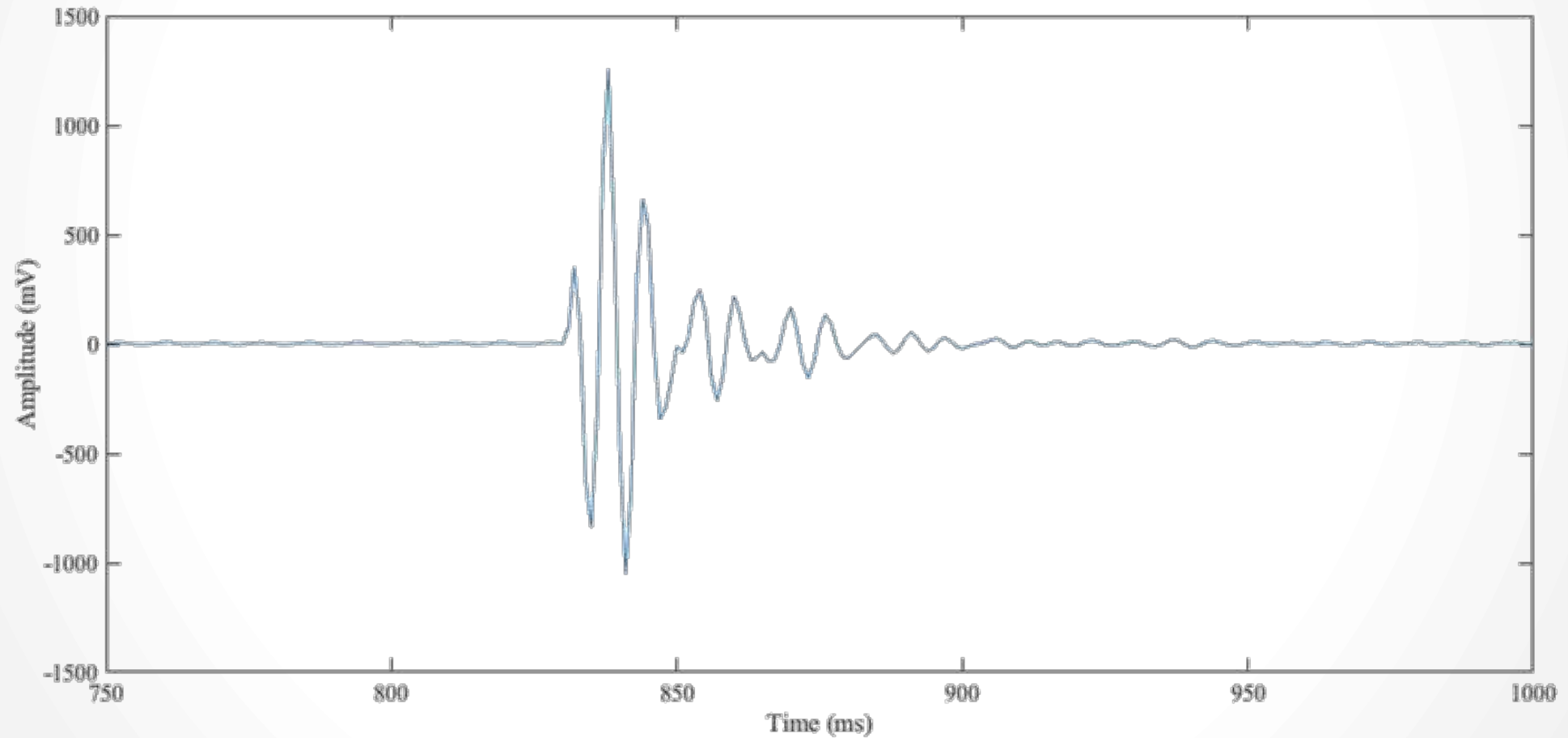


Accelerometer

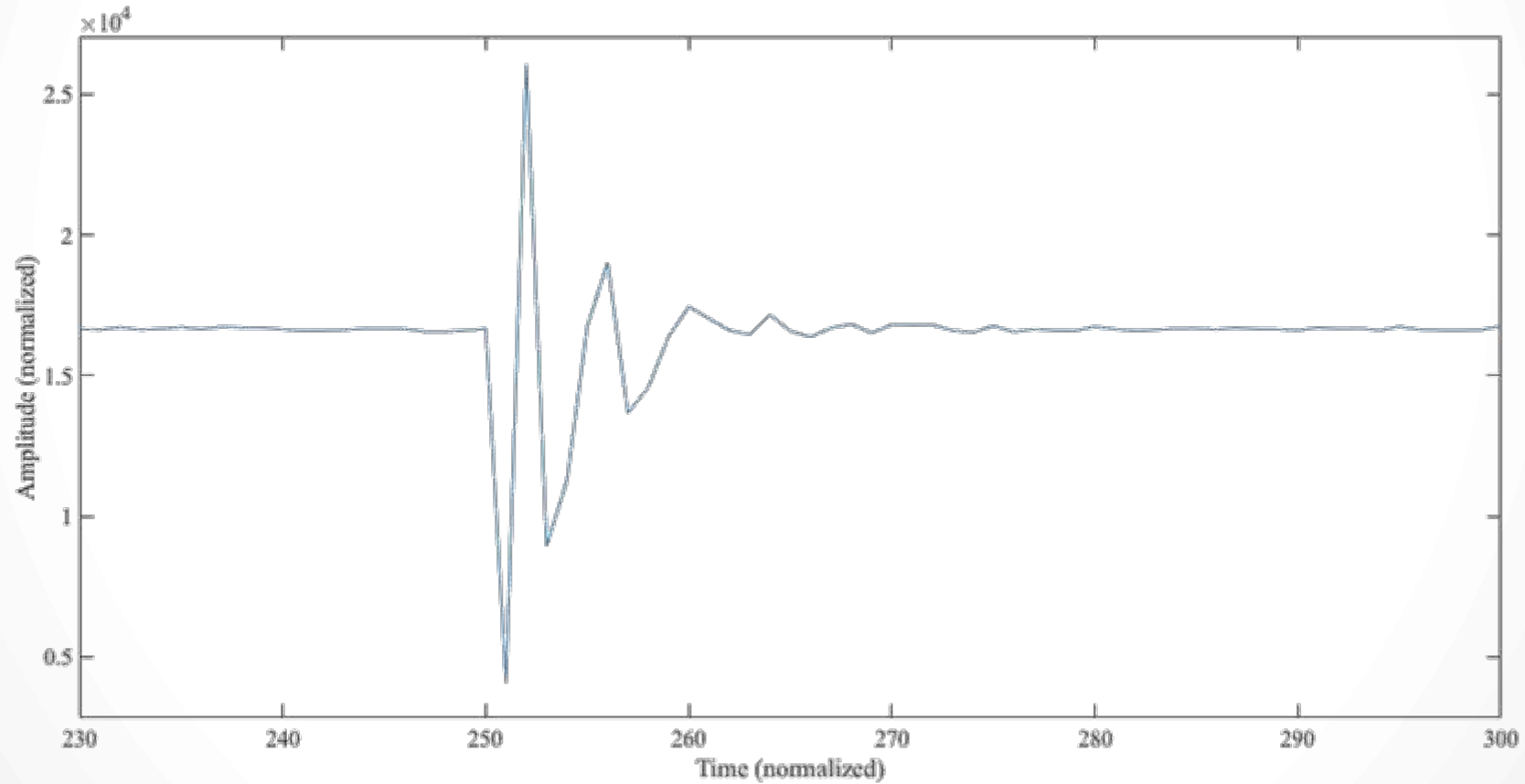
# Coil Based Geophone Impulse Response



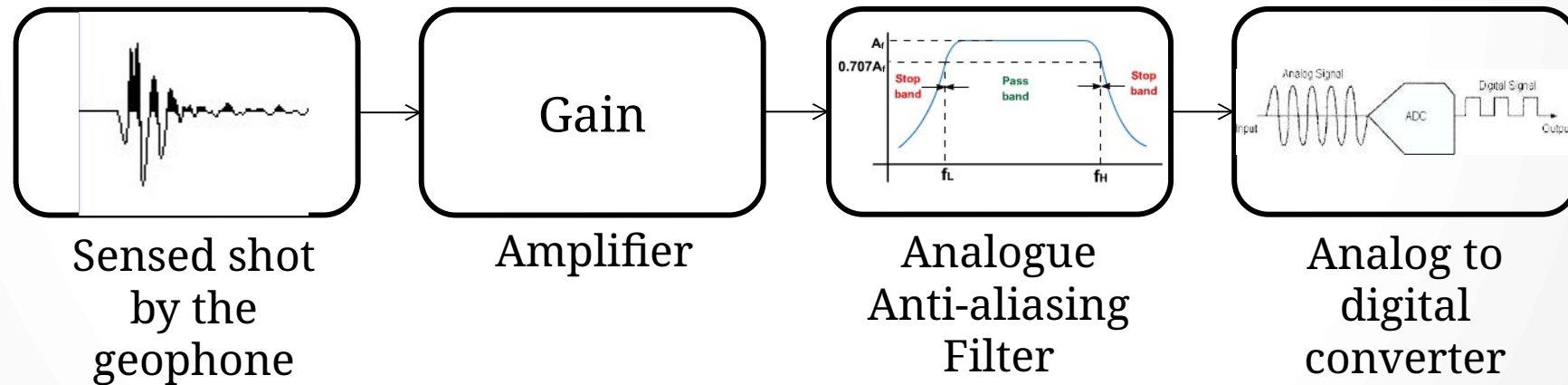
# Piezoelectric Impulse Response



# Accelerometer Impulse Response



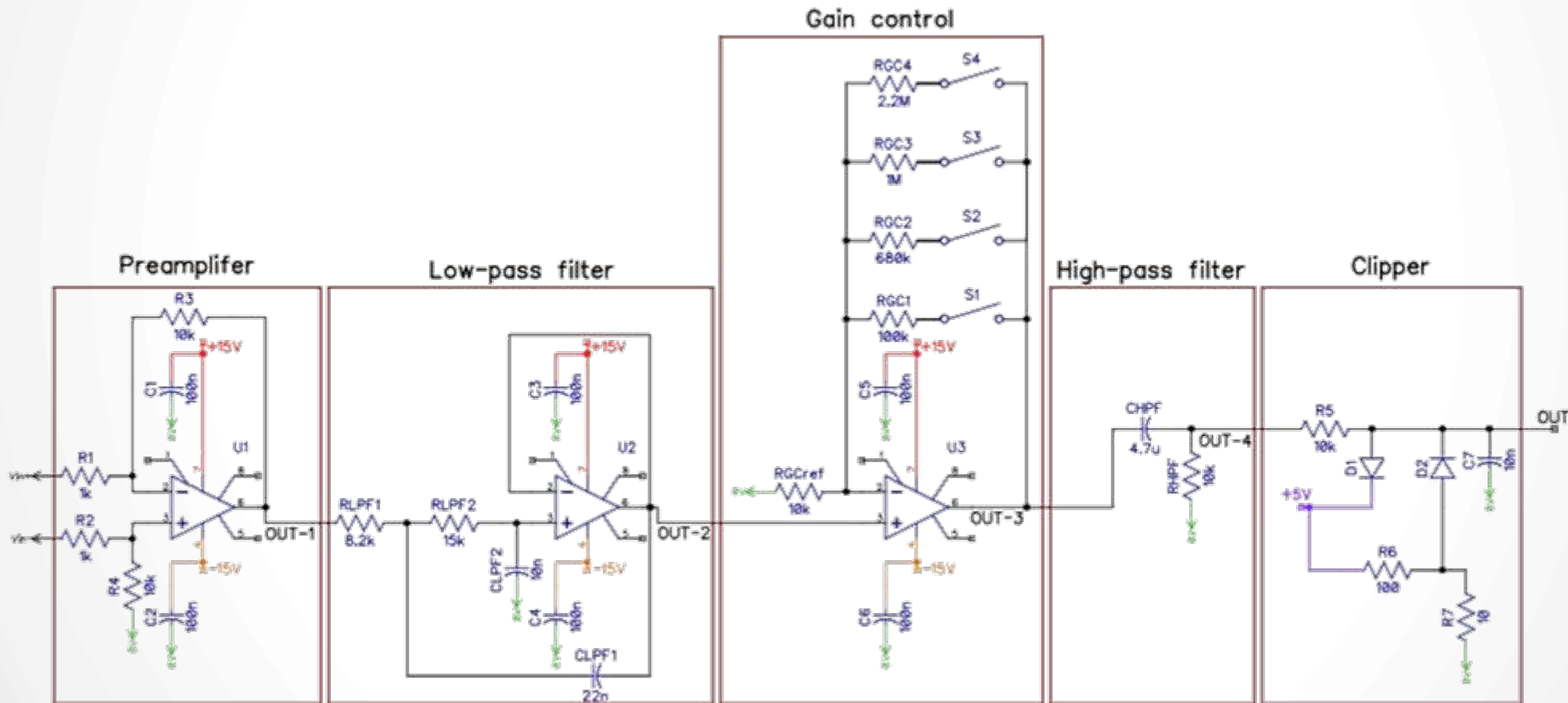
# Signal conditioning phase



# Signal conditioning circuit (S.C.C.)

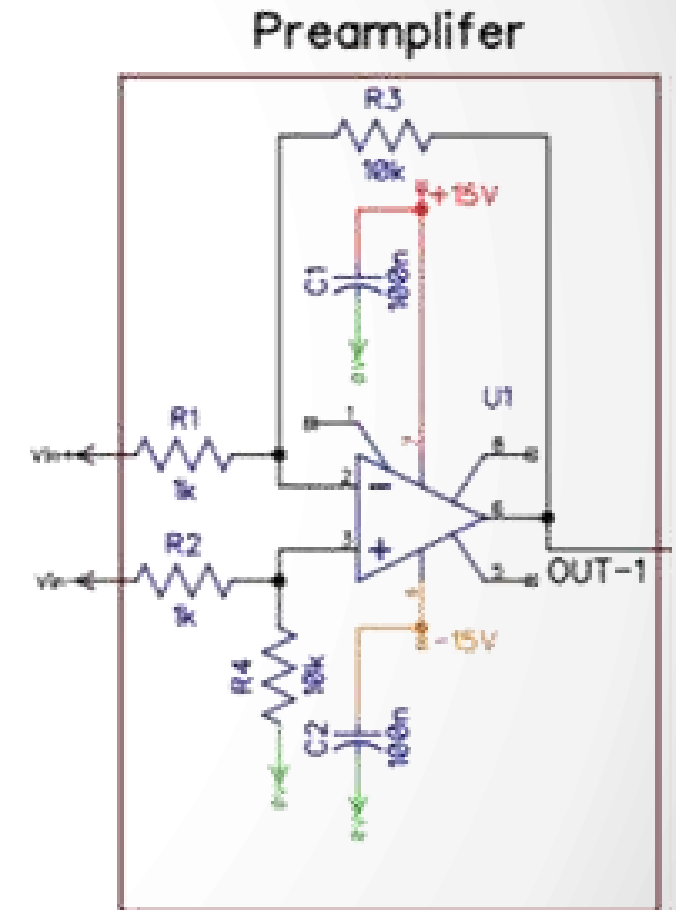
- Why S.C.C?
- Provides valid input-to-output data as defined by the system requirements over a given frequency range.
- S.C.C. main parts:
  1. Gain/signal conditioning cell
  2. An anti-aliasing filter
  3. An analog-to-digital converter (ADC)

# Signal conditioning circuit (S.C.C.)



# Signal conditioning circuit (S.C.C.)

- Preamplifier
- $|Gain| = \frac{R_3}{R_1} = 10 \text{ } v/v = 20 \text{ dB}$
- Used Op-amp (NE5532)
- Cost (1.025 \$ = 3.84 SR)



# Signal conditioning circuit (S.C.C.)

- Low-pass filter

$$f_{LPF} = \frac{1}{2\pi\sqrt{R_{LPF1}R_{LPF2}C_{LPF1}C_{LPF2}}} \approx 968 \text{ Hz}$$

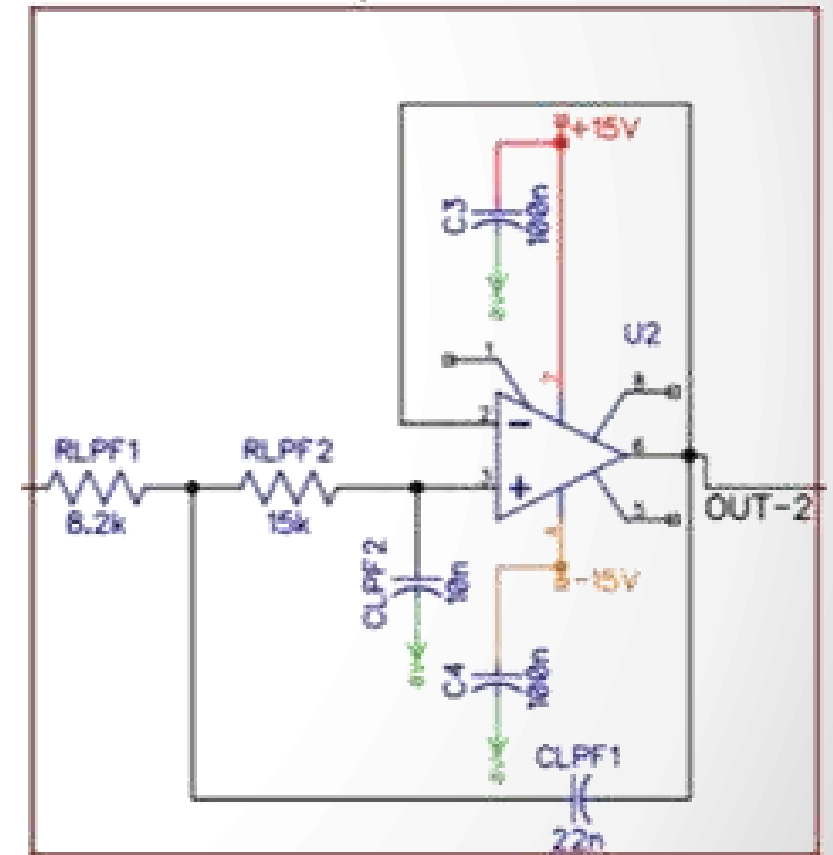
$$Q = \frac{1}{\omega_o*(R_{LPF1}C_{LPF1}+R_{LPF2}C_{LPF1})}$$

$$\text{Gain} = K = 1 \text{ v/v} = 0 \text{ dB}$$

$$H(s) = \frac{K}{s^2 + \frac{\omega_o}{Q}s + \omega_o^2}$$

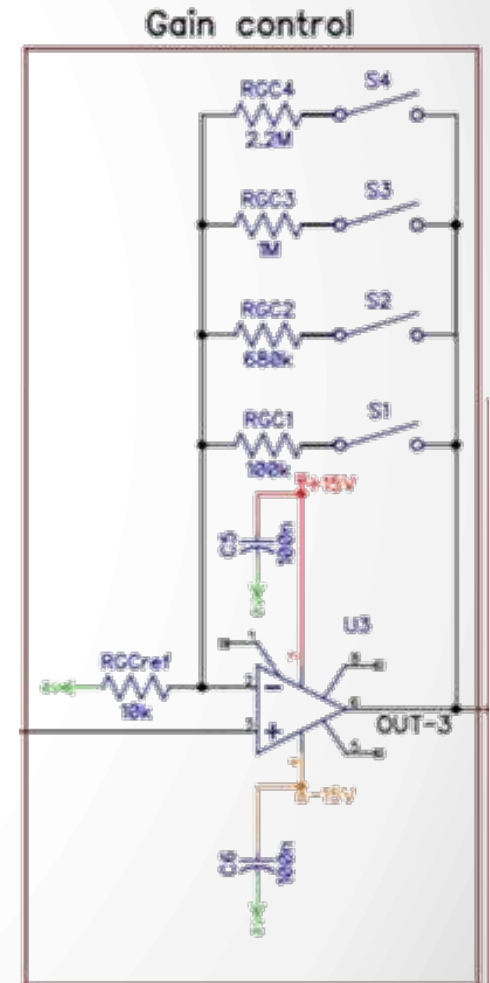
- Used Op-amp (OP07)
- Cost (0.894 \$ = 3.35 SR)

Low-pass filter



# Signal conditioning circuit (S.C.C.)

- Gain Control
- $\text{Gain} = G_n = 1 + \frac{R_{GC(n)}}{R_{GCref}}$
- $G_1 = 11 \text{ } v/v$
- $G_2 = 69 \text{ } v/v$
- $G_3 = 101 \text{ } v/v$
- $G_4 = 221 \text{ } v/v$
- Used Op-amp (OP07)
- Cost (0.894 \$ = 3.35 SR)

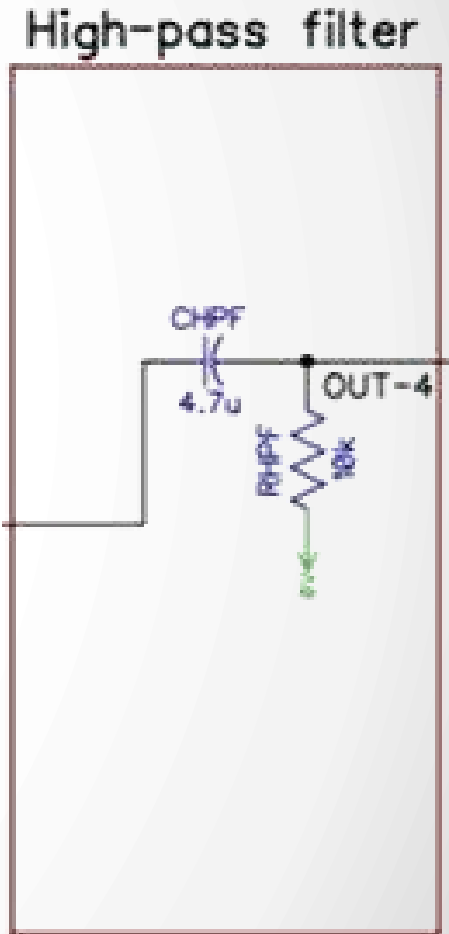


# Signal conditioning circuit (S.C.C.)

- High-pass filter

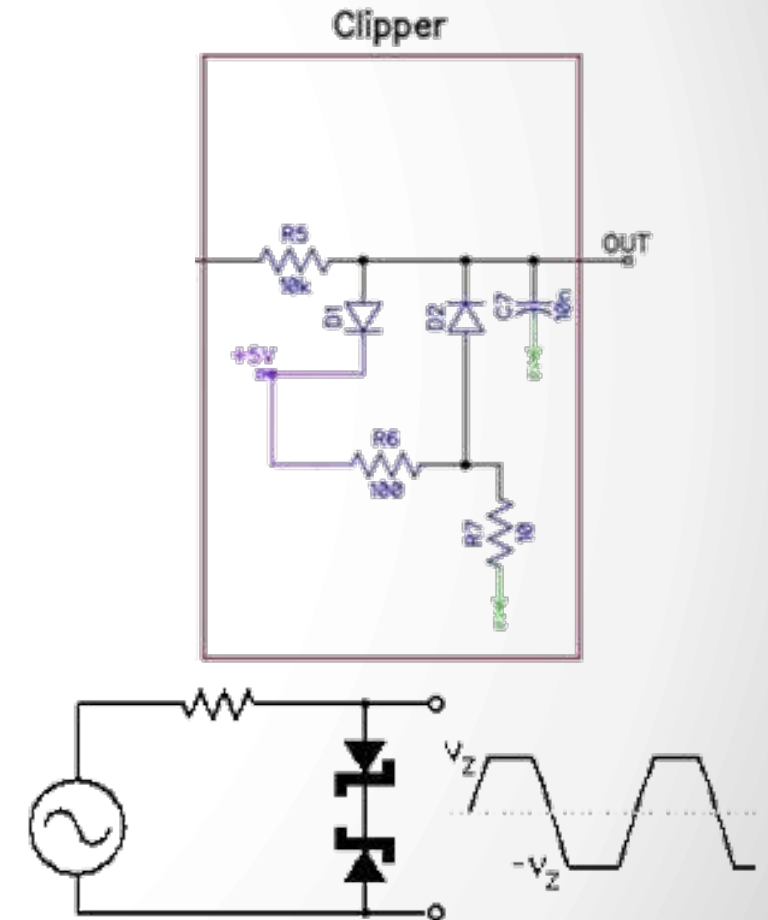
- $f_{HPF} = \frac{1}{2\pi R_{HPF} C_{LPF}} \approx 3.386 \text{ Hz}$

- $H(s) = \frac{\omega_o}{s + \omega_o}$

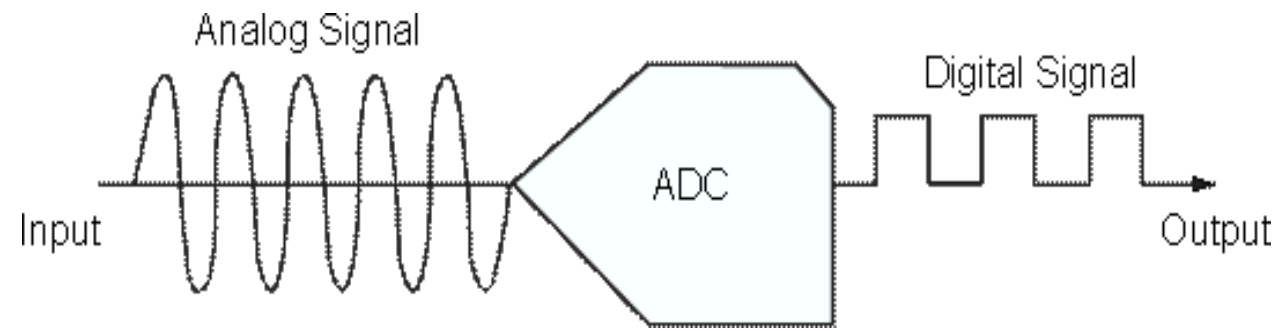


# Signal conditioning circuit (S.C.C.)

- Clipper (Limiter)
- *Voltage boundaries:  $-2.5\text{ V}$  to  $2.5\text{ V}$*



# Analog to Digital Conversion (ADC)



# Analog to Digital Converters (ADCs)

| Type           | Advantages                                                                                                                                    | Disadvantages                                                                                                                                                                                                                                   |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Flash          | <ul style="list-style-type: none"> <li>➤ Fast (almost instantaneous).</li> <li>➤ Simplicity.</li> </ul>                                       | <ul style="list-style-type: none"> <li>➤ Requires huge number of parts (exponentially proportional to the number of bits).</li> <li>➤ Low resolution (mainly caused by the mismatch between parts)</li> <li>➤ High power consumption</li> </ul> |
| SAR            | <ul style="list-style-type: none"> <li>➤ Constant power dissipation versus sampling rate.</li> <li>➤ High resolution.</li> </ul>              | <ul style="list-style-type: none"> <li>➤ Slow conversion.</li> <li>➤ Easily subjected to noise causing inaccurate readings.</li> </ul>                                                                                                          |
| Dual Slope     | <ul style="list-style-type: none"> <li>➤ High accuracy.</li> <li>➤ Immune to noise (coined to the conversion differential nature).</li> </ul> | <ul style="list-style-type: none"> <li>➤ Slow conversion</li> <li>➤ Expensive.</li> <li>➤ Accuracy may be affected by improper external connections.</li> </ul>                                                                                 |
| $\Delta\Sigma$ | <ul style="list-style-type: none"> <li>➤ High resolution.</li> <li>➤ Low power consumption.</li> <li>➤ Low cost.</li> </ul>                   | <ul style="list-style-type: none"> <li>➤ Not practical for high frequency applications due to oversampling.</li> </ul>                                                                                                                          |

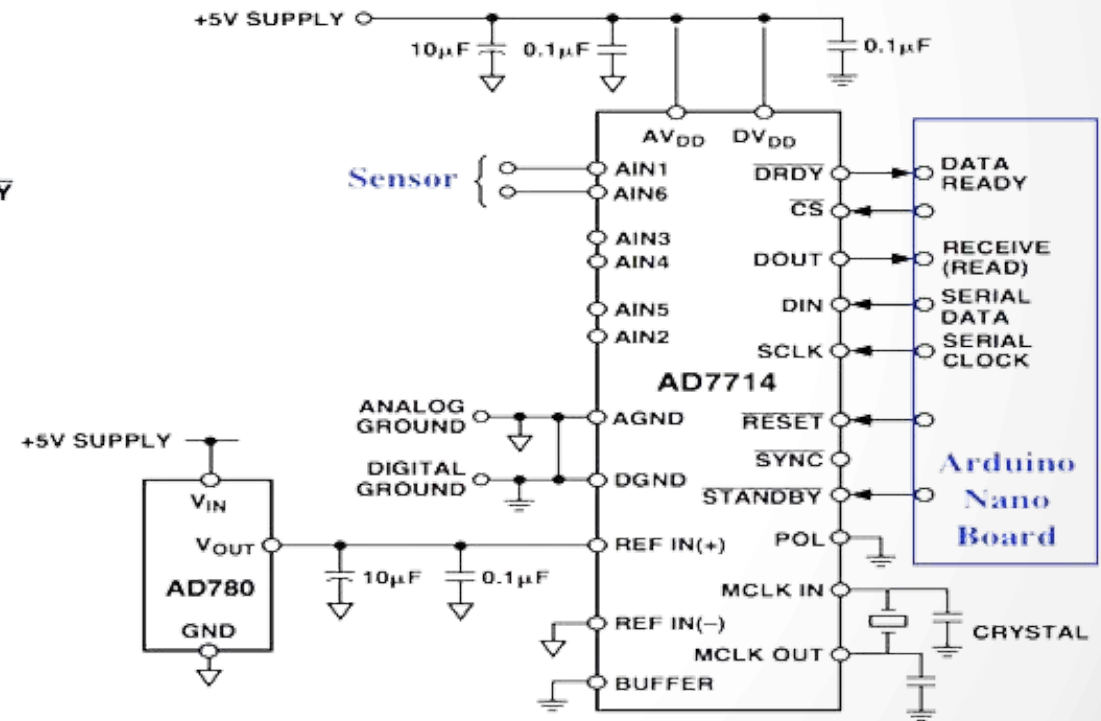
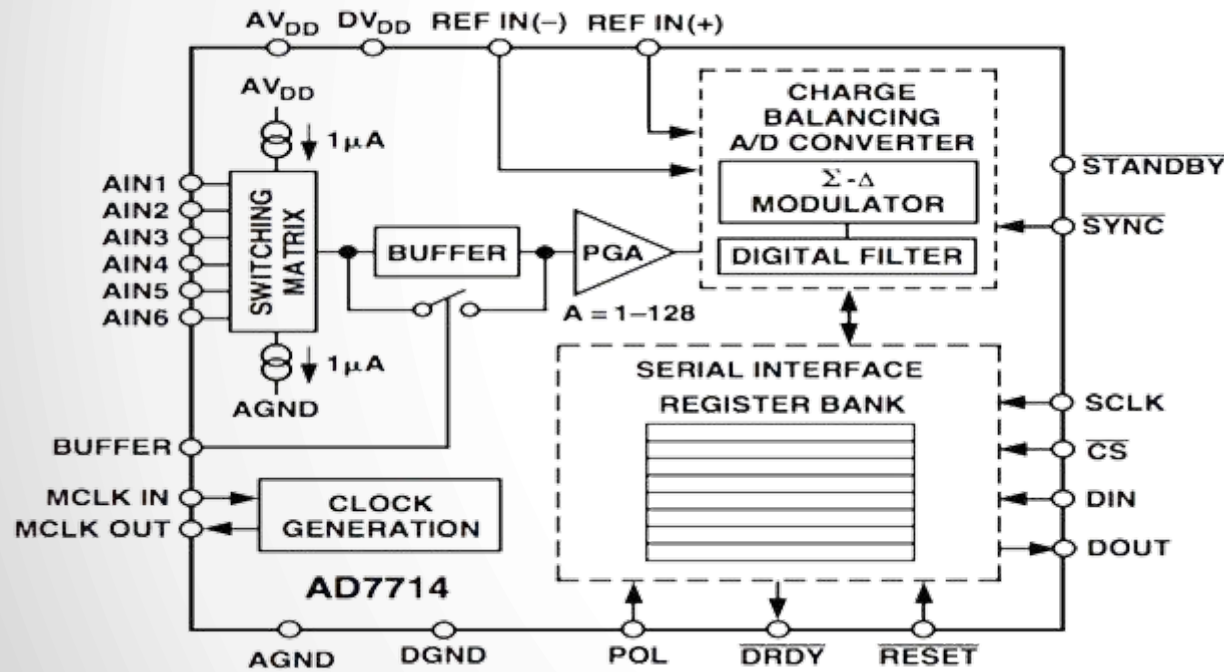
# Why Delta-Sigma ADC ?

- High resolution.
- Suitable for low frequencies (seismic data typical sampling  $\approx 500Hz$ ).
- Low cost and power consumption.

# Used Chip: AD7714 (19.73\$ $\approx$ 74 SR)

- Desirable features:
  - Low power consumption ( $\approx 500\mu A$ ) and dissipation (standby  $\approx 15\mu W$ ).
  - Bipolar dynamic range =  $\pm 2.5V$  (LSB  $\approx 0.2\mu V$ ).
  - Programmable gain (1~128) and low pass filter (4.8Hz~1.01kHz).
  - SPI interfacing.

# Functional Diagram & Circuit Connection



# Chip Registers

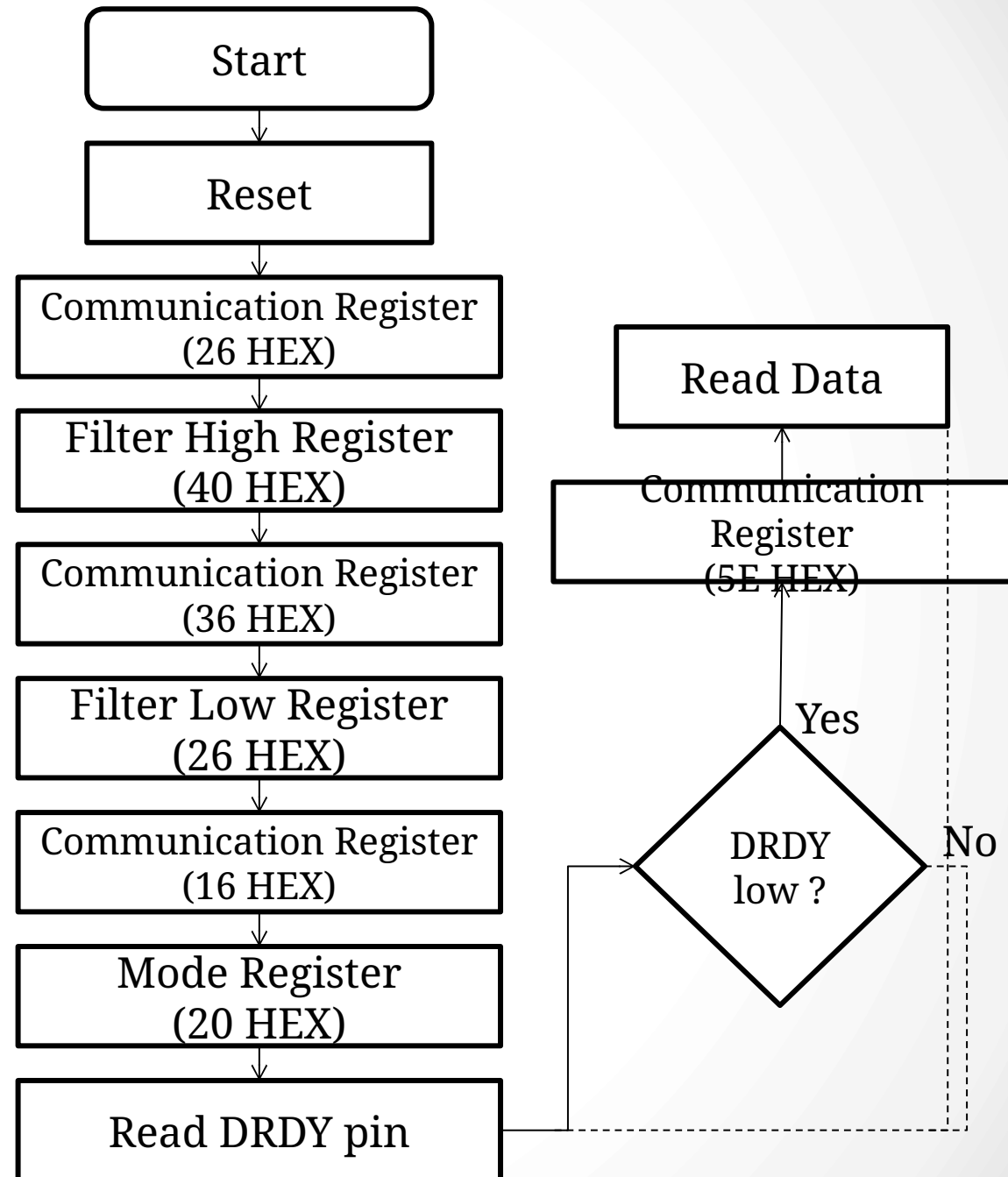
| RS2 | RS1 | RS0 | Register                        | Register Size      |
|-----|-----|-----|---------------------------------|--------------------|
| 0   | 0   | 0   | Communications Register         | 8 Bits             |
| 0   | 0   | 1   | Mode Register                   | 8 Bits             |
| 0   | 1   | 0   | Filter High Register            | 8 Bits             |
| 0   | 1   | 1   | Filter Low Register             | 8 Bits             |
| 1   | 0   | 0   | Test Register                   | 8 Bits             |
| 1   | 0   | 1   | Data Register                   | 16 Bits or 24 Bits |
| 1   | 1   | 0   | Zero-Scale Calibration Register | 24 Bits            |
| 1   | 1   | 1   | Full-Scale Calibration Register | 24 Bits            |

## Communications Register

|                             |     |     |     |                                |     |     |     |
|-----------------------------|-----|-----|-----|--------------------------------|-----|-----|-----|
| 0/ $\overline{\text{DRDY}}$ | RS2 | RS1 | RS0 | $\text{R}/\overline{\text{W}}$ | CH2 | CH1 | CH0 |
|-----------------------------|-----|-----|-----|--------------------------------|-----|-----|-----|

# Configuration

- Differential input (AIN1 & AIN6).
- Initial self-calibration state when the chip is powered on.
- Sampling frequency of 500Hz.
- Gain zero (was set to x8 whenever the Pre-ADC gain was not sufficient).



# Wireless Communication

- Two options were available:

1- nRF24L01+ RF Module

2- Digi XBee S2C RF Module

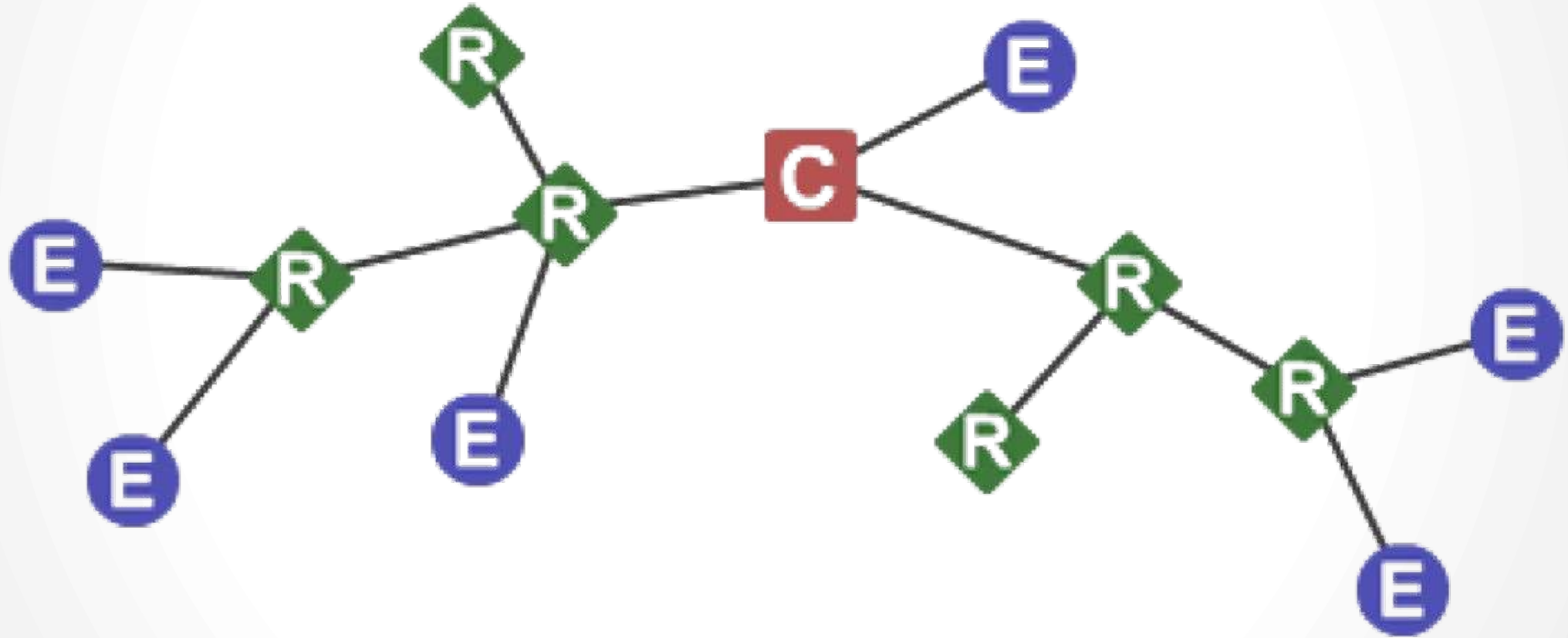
| Module              | nRF24L01+ RF Module                    | Digi XBee S2C RF Module |
|---------------------|----------------------------------------|-------------------------|
| Data Rate           | data rates of 250kbps, 1Mbps and 2Mbps | 250 kbps and 5Mbps      |
| Serial Interface    | SPI                                    | UART , SPI              |
| Operating Band      | ISM band                               | ISM band                |
| Communication Range | 100 m                                  | 1.2Km                   |

# Wireless Communication

- Features Xbee S2C Zigbee was chosen for:
  - Larger line-of-sight transmission range, compared to nRF24L01+
  - Designed for mesh networking
  - It uses Zigbee standard that is based on IEEE 802.15.4 standard



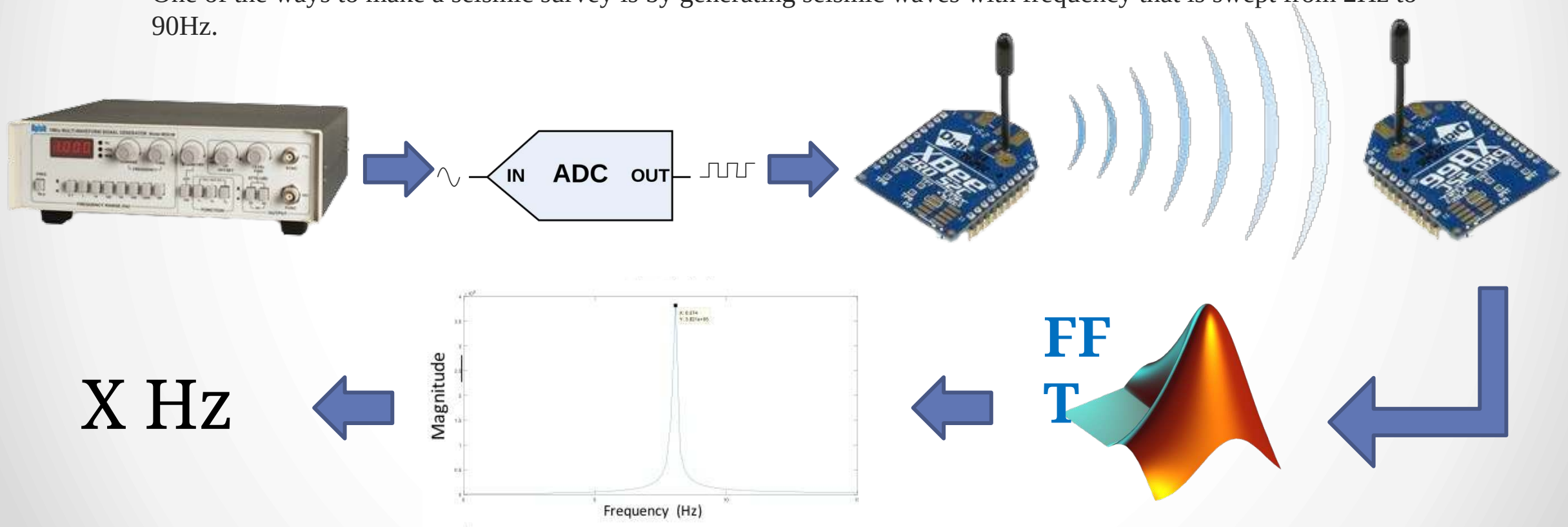
# Wireless Communication



# Wireless Communication

- Xbee S2C Zigbee Verification Test

One of the ways to make a seismic survey is by generating seismic waves with frequency that is swept from 2Hz to 90Hz.



# Wireless Communication

| Wirelessly sent frequency | Corresponding FFT peak frequency | Wirelessly sent frequency | Corresponding FFT peak frequency |
|---------------------------|----------------------------------|---------------------------|----------------------------------|
| 2 Hz                      | 2.0 Hz                           | 35 Hz                     | 35.2 Hz                          |
| 5 Hz                      | 5.0 Hz                           | 50 Hz                     | 50.1 Hz                          |
| 8 Hz                      | 8.1 Hz                           | 60 Hz                     | 60.2 Hz                          |
| 10 Hz                     | 10.0 Hz                          | 70 Hz                     | 70.3 Hz                          |
| 15 Hz                     | 15.1 Hz                          | 80 Hz                     | 80.2 Hz                          |
| 20 Hz                     | 20.1 Hz                          | 90 Hz                     | 90.3 Hz                          |
| 25 Hz                     | 25.2 Hz                          | ---                       | ---                              |

# Signal Processing Unit

- The one responsible for:
  - Controlling and managing.
  - Receiving Data from the ADC.
  - Sending Data through the Wireless Communication Unit.

# Signal Processing Unit

- Components Selection:
  - 2 Different Category of Alternative

Signal Processing

FPGA

Microcontroller

**REJECTED**

**ACCEPTED**

# Controlling Unit

- For Microcontrollers choice 2 different alternatives were found

Signal Processing

ATtiny85

AT89s8253

ATmega328P

**REJECTED**

**ACCEPTED**

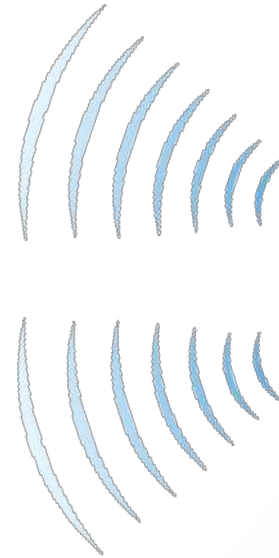
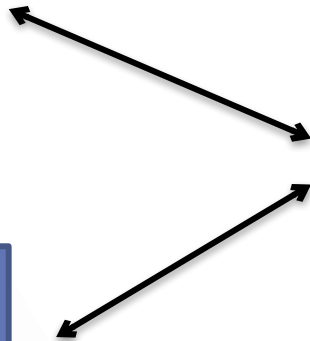
**ACCEPTED**

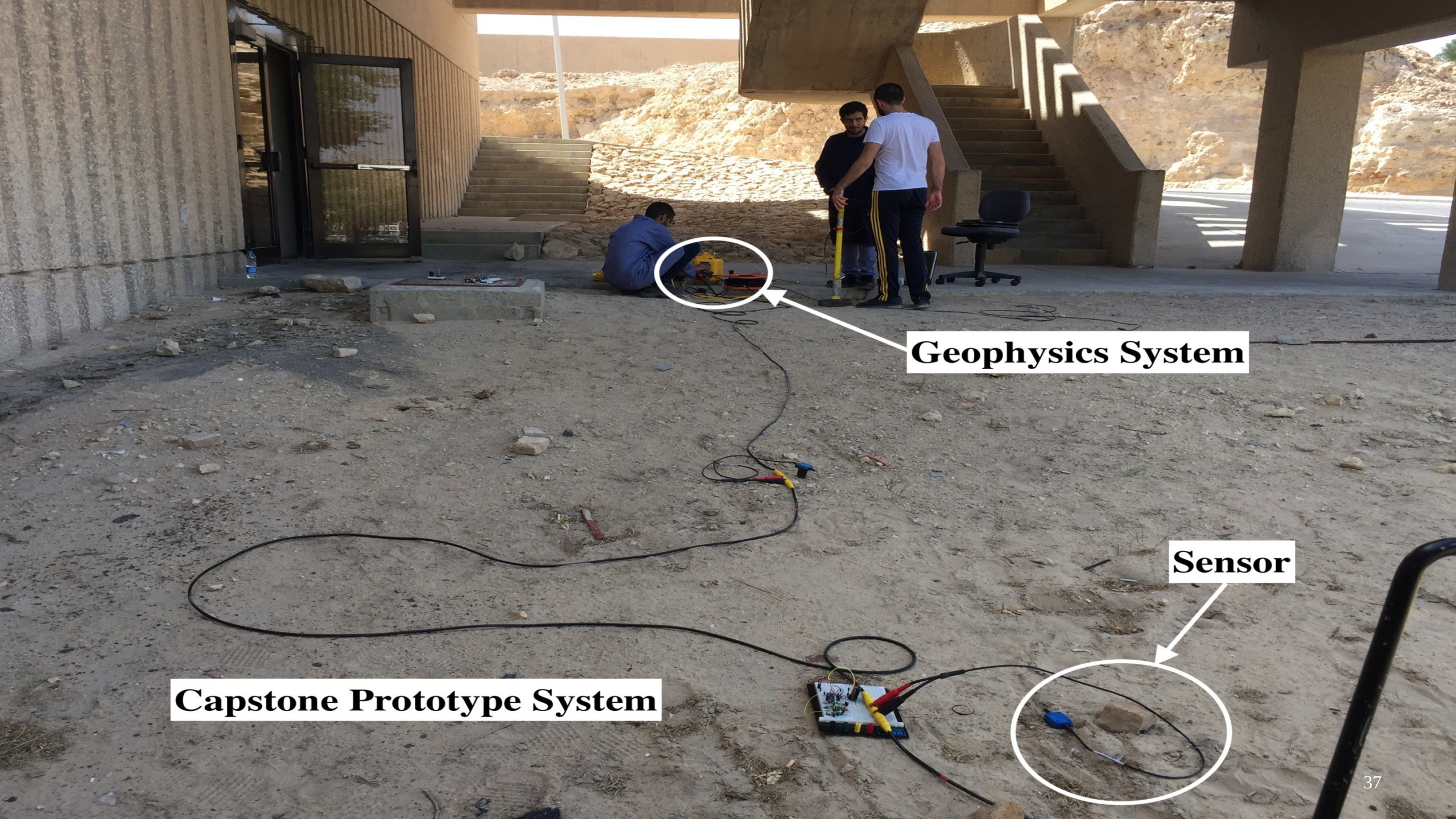
# Overall Verification

- Hummer experiment was performed at building 26.
- Experiment Setup:



The provided  
prototype





**Geophysics System**

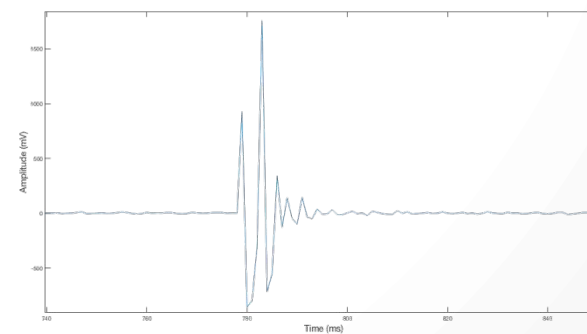
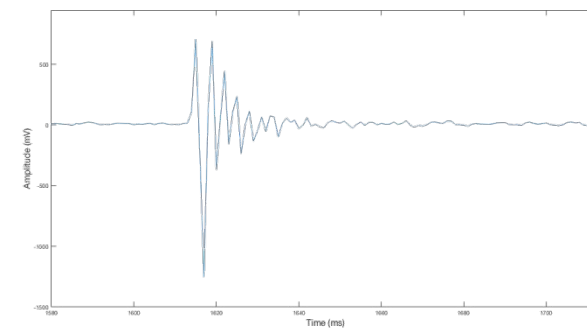
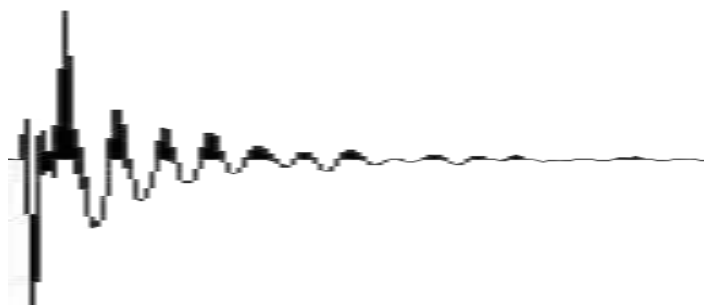
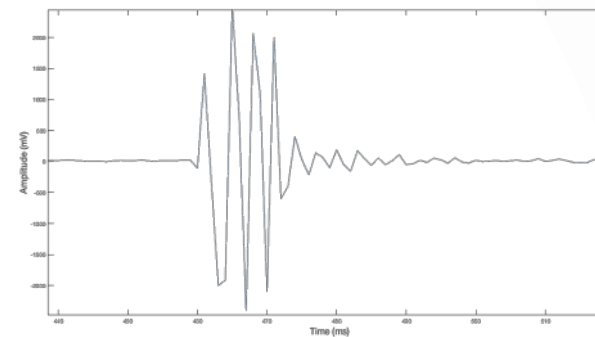
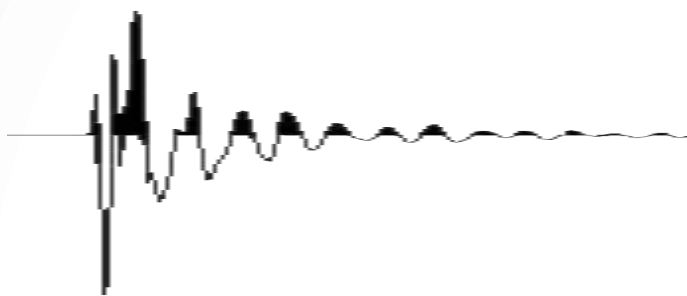
**Sensor**

**Capstone Prototype System**





# Overall Verification



# Conclusions and Recommendations

# Concluding remarks

- Gained understanding of data acquisition systems in general.
- Sensors were investigated based on their frequency response.
- Delta Sigma ADC (24-bits) was programmed and configured properly.
- Successful Wireless communication between two nodes up to 50m.
- Final working prototype was soldered and can be easily used.

# Recommendations

- Sensing the seismic signals with more than one sensor at a time.
- Including multiple nodes.
- Adjusting the Pre-ADC signal conditioning parameters.
- Find alternatives to the XBee adapter module.
- Writing the algorithm in another programming language.

## BPF + ADC

